When Language Models Quote Uncertainty: Separating Reasoning Errors from Measurement Noise

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Abstract

[Abstract to be written after the results sections.]

1 Introduction

[To be written. Outline: the question; why a benchmark with known $\sigma^\star$; the three findings, one per panel of Figure 1; contributions stated narrowly (docs/paper_outline.md, section 1).]

2 Related work

[To be written after the references in docs/paper_outline.md, section 2, are verified (R5-10).]

3 Setup

3.1 A generator of physics problems with known uncertainty

The benchmark is a procedural generator of multi-step physics word problems (generator version 2.0.0). It has 3 formula families (kinematics, dynamics, energy), each one physical scenario with named, dimensioned quantities and a set of relations written in SymPy [13]: 12, 15 and 10 relations respectively. A problem is a linear chain through its family's formula graph. The generator samples a consistent world (basis quantities log-uniform within physical ranges, derived quantities checked against plausibility bounds), picks a chain, shows the chain's inputs (the *givens*) and asks for its output (the *target*). The depth of a chain, from one step up to 5, is its difficulty dial. Depth is honest: the target's shortest derivation from the givens, over all relations of the family, has exactly the chain's length, and no given can be derived from the others. Depth 5 is never used for training. It tests extrapolation in chain length within the same families, not transfer to new physical laws.

Each problem's text is randomized: phrasing templates, display units with SI prefixes and non-SI units, and 0–2 distractor quantities that the answer does not depend on. Chains whose output amplifies relative errors in a given by more than a factor of 5 (the largest $|\partial \ln(\text{output})/\partial \ln(\text{given})|$) are rejected. The ground truth is computed in SI units with dimension checks at every step, then converted to the requested unit. Every problem is reproducible from its generator version and seed.

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3.2 Measurement noise and the true uncertainty $\sigma^\star$

In the measurement-noise mode, each noisy given is shown as a measurement $x_{\text{meas}} = x_{\text{true}}(1+e)$, $e \sim \mathcal{N}(0, s^2)$, with 5 significant figures, so that rounding is negligible against the noise, and a stated relative uncertainty "$\pm\, s$"; a value shown without one is exact. The truth is the noise-free chain evaluated on the unrounded true values. A problem without noise shows 3 significant figures, and its truth is computed from the displayed values. The pair sets of Section 3.5 use the levels 1 %, 2 %, 5 %, and the training data of Section 4.4 draws a level for each given independently from 0 %, 1 %, 2 %, 5 %, where a level of zero means an exact given.

Because the generator knows the measurement model, it knows each problem's correct uncertainty. Given the displayed values, the clean value of each noisy given is drawn as $x = x_{\text{meas}}/(1+e')$, $e' \sim \mathcal{N}(0, s^2)$, with exact givens and constants held fixed; 4,000 such draws are pushed through the chain, as in Monte Carlo propagation of distributions [11]. The target $\sigma^\star$ is the scale of the Gaussian in log space, centred at $\ln f(x_{\text{meas}})$, that minimizes the expected CRPS against these draws:

$$\sigma^\star = \arg\min_{\sigma>0} \; \mathbb{E}_x\Big[\mathrm{CRPS}\Big(\mathcal{N}(\ln f(x_{\text{meas}}), \sigma^2),\, \ln f(x)\Big)\Big]\,, \tag{1}$$

where f is the chain and $f(x_{\text{meas}})$ is what a correct solver computes from the displayed values. The standard deviation of the log draws and the first-order (delta-method) value are stored as well.

Before any model was evaluated on noisy problems, the generator had to pass a self-consistency gate: intervals $\ln f(x_{\text{meas}}) \pm k\sigma^\star$ must cover the clean truth at their nominal rates. On the final run, with 10,005 noisy problems in 15 depth-by-level cells of 667 each, pooled coverage was 68.32 % at $k = 1$ (nominal 68.27 %, Kupiec [12] $p = 0.92$) and 95.71 % at $k = 2$ (nominal 95.45 %, $p = 0.20$). No cell was rejected after Holm correction, and the probability integral transform of the truth was uniform (Kolmogorov–Smirnov $p = 0.98$). The 500 noise-free control problems were exact. The gate failed twice before this run. The first run failed on one cell at the highest noise level, after which the noise levels were capped; that cap was chosen after seeing the result. The second failed only on a rule that, as written, tolerated no cell outside its interval; it was replaced by a miss-count test before the final run. Both changes are listed in Table 2.

3.3 Output format and verifier

A completion may reason freely, then reports the requested intermediate values in a `<steps>` block and the answer as `<answer>`$\mu \pm \sigma$ *unit*`</answer>`. A completion that does not parse, or whose unit has the wrong dimension, scores the rule's floor (-5); an answer in another unit of the right dimension is converted.

The scoring rule, $\mathrm{crps}_{\log}$, scores the log of the truth y against $\mathcal{N}(\ln\mu, (\sigma/\mu)^2)$:

$$\mathrm{crps}_{\log}(\mu, \sigma; y) = -\,s\left[z\,(2\Phi(z)-1) + 2\varphi(z) - \tfrac{1}{\sqrt{\pi}}\right], \qquad z = \frac{\ln y - \ln\mu}{s}, \quad s = \max(\sigma/\mu,\ \epsilon)\,, \tag{2}$$

the Gaussian CRPS in closed form [6], clipped below at the floor. It is strictly proper for log-normal beliefs [5] and scale-free. The relative CRPS of the original specification divides by the truth, which reweights outcomes and makes it only approximately proper; it remains an ablation option. The floor ϵ is 1×10^{-6} (verifier 1.1.0): an earlier, larger floor made the reward flat in σ exactly where an overconfident policy needed a gradient. During reinforcement learning the reward adds 0.2 times the share of requested intermediates within 2 % of their values; this weight is checked to be too small to ever outweigh the final score. An answer is *accurate* if it is within 1 % of the truth.

On noisy problems, an answer is *computation-correct* if $|\ln \mu - \ln f(x_{\text{meas}})| \leq 1 \times 10^{-3}$, that is, if it is what a correct solver computes from the displayed values. Comparisons of a stated σ with $\sigma^\star$ that require a correct answer use this subset.

3.4 The oracle-centred expected score

A single realized truth says little about whether a stated σ is right: a σ that is 2 times too small scores better than $\sigma^\star$ against the realized truth on 44 % of problems. Because the generator knows the predictive distribution, we also score a stated relative σ by its *expected* $\text{crps}_{\log}$ over 4,000 fresh predictive draws, centred at $f(x_{\text{meas}})$. Per-problem differences between sources then have 5–23 times less variance than the realized ones. Expected-score differences are reported with their intervals and, where useful, as a share of the gap between the noise heuristic RSS (Section 4.5) and $\sigma^\star$, which is 0.00296 on the frozen confirmatory set.

3.5 Counterfactual pair sets

A model can state a plausible σ without propagating anything, by learning which templates, depths and stated percentages go with which uncertainty. To separate propagation from such a lookup, we evaluate on matched pairs of problems that differ in exactly one respect (Table 1). The statistic is the within-pair slope: the ordinary least-squares slope, with an intercept, of $\Delta \ln \sigma_{\text{stated}}$ on $\Delta \ln \sigma^\star$ across pairs, where Δ is the difference between a pair's two members. An exact propagator has slope one; a reader of features that do not differ within a pair has slope zero.

A6 (value changes). One chain template, two worlds: of 24 candidate worlds, the two with the lowest and highest delta-method $\sigma^\star$, kept if their ratio is at least 2.0. Template, phrasing, noise level and the standardized measurement errors are shared. Power-law chains cannot form A6 pairs, since their relative $\sigma^\star$ does not depend on the values.

A6b (invariance). Power-law templates only, two worlds whose targets differ by a factor of at least 2.0, with relative $\sigma^\star$ checked equal by Monte Carlo (to within 1×10^{-3} in log). Here the slope is taken on $\Delta \ln y$ and must be zero: a model whose relative σ grows with the size of the answer fails it.

A6c (noise allocation). One chain, one world and identical displayed values. The two members differ only in which given carries the higher of two noise levels: in one member it sits where it moves $\sigma^\star$ least, in the other where it moves it most (ratio at least 1.5). Both members show the same multiset of percentages, so no reader of the problem's surface can tell them apart.

Each development set has 15 depth-by-level cells of 34 pairs. The frozen confirmatory set (Section 5.2) repeats the three constructions with fresh seeds, 1,020 problems each.

Jacobians outside the training range (the J set). A model trained on a chain might have learned that chain's typical sensitivities rather than computing them. The J set therefore uses only chains that occur in the training data, and moves them to Jacobians they never had there. For each of the 1,149 chains in the 3,000 training problems we record the range of every given's log-sensitivity $\partial \ln f / \partial \ln x$ at the displayed values. A J world has at least one given whose sensitivity lies outside that range by at least 0.1, while every displayed and computed value stays inside the range that quantity had in training; the first such world of up to 96 draws is used. *J sensitivity pairs* apply the A6c construction to J worlds. *J invariance pairs* pair a J world with a copy rescaled along a dimensional direction that leaves every physical constant unchanged, which keeps every

Table 1: The counterfactual pair sets and what each isolates.

set	changes within a pair	held fixed	regressor	ideal	problems
A6	the world (values); same template	template, phrasing, level, z	$\Delta \ln \sigma^\star$	1	1,020 / 1,020
A6b	the target's magnitude ($\geq \times 2$); power-law template	relative $\sigma^\star$ (MC-checked)	$\Delta \ln y$	0	1,020 / 1,020
A6c	which given carries the high level	displayed values, level multiset	$\Delta \ln \sigma^\star$	1	1,020 / 1,020
J sensitivity	as A6c, on a Jacobian outside the training range	as A6c	$\Delta \ln \sigma^\star$	1	816
J invariance	a dimensional rescaling (target $\times 2$–8)	every log-sensitivity, so relative $\sigma^\star$	$\Delta \ln y$	0	816

Ideal: the within-pair OLS slope of $\Delta \ln$(stated σ) on the regressor for an exact propagator; a reader of surface features that are identical within a pair scores 0 on A6c and J. Problems: development / frozen confirmatory set (A6, A6b, A6c; D113), or the frozen J set (D147). Sources: the sets' manifests.

log-sensitivity, and so the relative $\sigma^\star$, identical while the target changes by a factor of 2 to 8; A6b cannot be used here because its power-law Jacobians are constant. The J set has 816 sensitivity and 816 invariance problems at the trained depths (up to 4). Across the sensitivity pairs, the most out-of-range sensitivity lies a median of 0.38 outside its chain's range, and the median chain was seen 3 times in training.

4 Methods

4.1 Model and training infrastructure

All models are Qwen2.5-Math-1.5B-Instruct [17], loaded with 4-bit NF4 quantization [3] from a pinned snapshot and trained with LoRA adapters [10] of rank 16 ($\alpha = 16$) on all 7 attention and MLP projections, using Unsloth [8] and TRL [16]. Adapters are evaluated as trained: unmerged, on the same quantized base. An early evaluation path merged reinforcement-learning adapters into a bf16 copy of the model, which erased most of the update; those results are superseded and kept only as labelled records. A cross-model pilot repeats the D-pos-v2 and B-v2 recipes of Section 4.4 on Llama-3.2-3B-Instruct [7].

4.2 Supervised warm-up

Reinforcement learning starts from a supervised warm-up on 3,000 generator solutions written in the output format: the needed SI values, one line per chain step with the substituted numbers, the `<steps>` block and the answer. The σ in these targets is a fixed 2 % of the value, and its tokens are masked from the loss, so the warm-up teaches the format and the computation but never a particular uncertainty. It runs for 30 minutes at learning rate 1×10^{-4}.

4.3 Reinforcement learning with a proper scoring rule

We train with GRPO [15]: for each of 4 problems per step the policy samples 8 completions at temperature 1.0, each completion is scored by the verifier (Section 3.3), and the reward is normalized

within its group. There is no KL penalty ($\beta = 0$); the learning rate is 5×10^{-6} and completions are capped at 512 tokens. Problems are drawn by a curriculum over (family, depth) cells that tracks each cell's pass rate p with an exponential moving average (rate 0.1) and samples cells in proportion to $p(1-p) + 0.02$, the weight that favours groups with reward variance. Runs are compared at a matched step (287).

Split credit. With sequence-level credit every token of a completion receives the same advantage, so the tokens of σ are rewarded for the answer's correctness rather than for their own calibration. Split credit decomposes the verifier's reward r into two parts that sum to it,

$$r_{\text{answer}} = \text{crps}_{\log}(\mu, \sigma_{\text{ref}}) + w\, r_{\text{steps}}, \qquad r_\sigma = \text{crps}_{\log}(\mu, \sigma) - \text{crps}_{\log}(\mu, \sigma_{\text{ref}}), \tag{3}$$

with a fixed reference σ_{ref} of 1 % of μ and the step weight w of Section 3.3. Each part is normalized within its group. The tokens of the σ number, located by parsing, receive the advantage of r_σ; every other token receives that of r_{answer}. A failed completion keeps its whole reward in r_{answer}. The verifier, the logged reward and every hyperparameter are those of the uniform-credit run; split credit and uniform credit are each trained with 3 seeds.

σ-only reinforcement learning (E1). To isolate credit assignment completely, E1 holds the answer fixed. Each training prefix is a warm-up model's own completion truncated right after the "±" of its answer, and the policy samples only the remaining σ suffix (at most 24 tokens), so that within a group only σ varies. E1 trains for 300 steps at learning rate 5×10^{-6}; the E1d pilot repeats it at 5×10^{-5}, and is treated as exploratory.

4.4 Explicit propagation: D-pos-v2 and its implicit control B-v2

Supervising $\sigma^\star$ directly, rewarding σ alone, and training an explicit propagation line on data without value-dependent derivatives all failed to make the stated σ propagate noise. The data of this section contain the behaviour. They reuse 3,000 training problems with mixed noise: each given's level is drawn independently from 0 %, 1 %, 2 %, 5 %, and 2,067 problems mix two or more noisy levels. Both arms see the same problems.

D-pos-v2 (explicit). After the solution, the target completion writes one line per noisy given x_i: the given's total log-sensitivity $S_i = \partial \ln f / \partial \ln x_i$ as a symbolic expression derived by the chain rule over the steps, the same expression with the solution's values substituted, and its value to 3 significant figures. A power-law term is stated as its exponent. A propagation line then combines the terms with the stated levels s_i,

$$\sigma_{\text{rel}} = \sqrt{\textstyle\sum_i (S_i\, s_i)^2} = X, \tag{4}$$

and the answer states $\sigma = X\,|\mu|$. Every token is trained, including the sensitivities, the line and σ. We read three values of σ from each D-pos-v2 completion: (a) the answer's relative σ; (b) the line's stated X; (c) the exact value of Equation (4) computed from the line's own written terms. Comparing them locates a failure in the propagation itself, the arithmetic of the line, or the copy into the answer.

B-v2 (implicit). The same problems and solutions with no sensitivity lines and no propagation line; $\sigma^\star$ itself is written after "±" and trained. B-v2 can learn propagation only implicitly.

Training and evaluation. Each arm trains a fresh adapter on the pinned base for 2 epochs (750 steps) at learning rate 1×10^{-4} and effective batch 8, with 3 seeds per arm. The seed changes the adapter's initialization and the data order, not the data. Evaluation is greedy, with up to 1,280 new tokens, enough for the longest derivations.

4.5 Reference uncertainties

Every stated σ is compared with three references that need no model. $\sigma^\star$ is the generator's own value (Equation (1)). RSS is the root sum of squares of the stated levels of the givens, $\sqrt{\sum_i s_i^2}$: correct for a product of first powers, and blind to the values. X1 is a shallow text model of $\sigma^\star$: gradient-boosted trees on the problem text's tokens, the stated levels and the displayed magnitudes, fitted on generator problems disjoint from every evaluation set. It is refit exactly and used only if it reproduces its published results. On the J set we add a fourth, *clamped* reference: the first-order σ with every sensitivity clipped into its chain's training range, which is what a model that propagates exactly but only within its training range would state.

4.6 Executing instead of predicting: MC-A and MC-B

MC-A estimates σ by re-asking the model. For each problem it builds 8 perturbed copies in which every noisy given is redrawn from the generator's own predictive distribution, $x = x_{\text{meas}}/(1 + sz)$ with $z \sim \mathcal{N}(0, 1)$, and shown in its displayed unit; B-v2 answers each copy. Its σ is the spread of $\ln \mu$ over the copies: the standard deviation (raw), or 1.4826 times the median absolute deviation over the copies whose solution steps match the unperturbed one (robust). The same perturbations pushed through the exact chain give an oracle that targets $\sigma^\star$.

MC-B asks the model for the computation instead of the answer, in the spirit of program-aided reasoning [1, 4]. The base model, without an adapter, writes a Python function `solve` whose parameters are the givens' own symbols; 2 worked examples from a disjoint seed range at depth 2 show the format, and generation is greedy with up to 640 new tokens. A sandbox checks the program's syntax tree (only the `math` module, no dynamic code), calls it on the displayed values and on MC-A's perturbations, and takes the spread of the logarithms of its outputs as σ. A failing program is classified as an interface failure (no program, wrong signature, rejected), a runtime failure, or a semantic failure (it runs but computes the wrong chain).

5 Evaluation protocol

5.1 Pre-registration and the decision record

The project's hypotheses were pre-registered in a file written before any result and never edited afterwards. Every experiment after that was specified in a dated entry of a public decision log, pushed to the repository before its data existed; the commit is the timestamp, and the few amendments made after a result are labelled (Section 5.4). The log has 150 entries. Each pre-registration states the hypothesis, the estimator, the decision rule and what would count as failure, and labels the analysis confirmatory or exploratory. Decisions taken during unattended runs were logged in the same way, with the alternatives considered. Results are recorded as numbered findings (39 so far), each citing its entry, and a negative result is reported with the same prominence as a positive one. Pre-registered hypotheses that were not run are declared as such rather than dropped. A chronological timeline of every pre-registration and amendment, with its status, accompanies the paper.

5.2 Frozen evaluation sets

The development pair sets of Section 3.5 were used to build and debug every analysis. Confirmatory claims rest on frozen sets built with fresh seeds after their analysis was specified: the R4-5 set (A6, A6b and A6c, 1,020 problems each) and the J set. Each frozen file has a SHA-256 manifest; the evaluation loader refuses a frozen file unless the run sets an explicit confirmatory flag, and regenerates every problem from its seed and compares it before any generation. No model saw a frozen set before the confirmatory run that its entry registered. While those runs were in progress, interim reports gave only operational counts (completions, truncation, parse failures, memory backoffs), and each confirmatory analysis ran once, after all of its evaluations existed.

Every completion is logged with its problem, verifier breakdown and provenance, and replaying a log regenerates each problem from its seed and re-scores the completion exactly. Every adapter carries a fingerprint recorded at the end of training, and an evaluation reloads it and checks the fingerprint before generating. Long evaluations write each finished batch to disk under a key of the model, adapter, prompts and generation settings, so an interrupted greedy evaluation resumes without changing any completion.

5.3 Statistics

Pair slopes. For each pair set the statistic is the within-pair OLS slope with an intercept (Section 3.5): of $\Delta \ln \sigma_{\text{stated}}$ on $\Delta \ln \sigma^{\star}$ for A6, A6c and the J sensitivity pairs, and on $\Delta \ln y$ for A6b and the J invariance pairs. A stated σ of zero enters at a relative floor of 1×10^{-6}. For an answer's σ, the noise-allocation slopes use pairs whose two members are both answered, and the invariance slopes use pairs whose two members are both computation-correct, so that a wrong answer cannot pass as invariance; values read from a propagation line use every pair.

Uncertainty over pairs and seeds. Confidence intervals come from a bootstrap over pairs with 10,000 resamples. With several seeds, each resample draws the pairs once and recomputes every seed's statistic on them; the estimate is the mean of the per-seed statistics [2]. One-sided p-values are the share of resamples at or below zero. Differences in expected score (Section 3.4) are averaged over seeds per problem and bootstrapped with pairs as clusters. Within-problem comparisons of correctness signals are clustered over problems. Neither pairs nor seeds that share problems are ever treated as independent.

The propagation criterion. A σ source propagates if three conditions hold: the noise-allocation slope is positive (C1: the lower end of its 90 % interval is above zero); it is at least 0.5, a practical threshold (C2); and the invariance slope is equivalent to zero, with its 90 % interval inside ± 0.3, a two one-sided tests criterion [14] (C3). The primary hypotheses of the confirmatory runs (P on the R4-5 set, P_J on the J set) apply this criterion to D-pos-v2's answer σ, with C1 replaced by a one-sided test that is Holm-adjusted within its family.

Multiplicity. Each confirmatory run fixes its family of tests in advance (K1–K8 on the R4-5 set, J1–J6 on the J set), and p-values are Holm-adjusted [9] within it at level 0.05. Subgroup estimates (by depth, chain type or distance from the training range) are reported with intervals but are outside the family, and no confirmatory claim rests on them. A score difference that is not significant is reported as not rejected, never as a tie.

5.4 Deviations

Table 2 lists every amendment to a pre-registered plan and every deviation that touched a registered analysis, with when it was made relative to the data it affects. Most were made before any affected result existed. Those made after a result are labelled as such wherever the result is reported.

Table 2: Amendments and deviations from the pre-registered plans.

entry	date	what	timing
D47	10-14 10:41	P's pre-registered sanity check failed; exploratory baselines added afterwards	amendment, **after** P's result
D49	10-14 11:54	Two post-hoc calibration baselines; RL calibration judged against the text-feature one	amendment, before RL calibration results
D50	10-14 11:59	Review round 1, A1–A8: two estimands; the σ^* definition; the generator self-consistency gate (A3); statistics; nested A5 baselines and family N …	amendment, before any result it affects
D54	10-14 12:43	Noise capped at 5%; 10% becomes a heavy-tail regime	amendment, **after** the gate failed (#9)
D57	10-14 12:56	A3's cell-share rule replaced by a miss-count test; the next rerun is FINAL	amendment, **after** #11, before the final rerun
D61	10-16 14:34	A6 slope labels ("partial / full propagation")	amendment, before any A6 model result
D62	10-16 19:12	Split-credit seeds 1 and 2; the seed-mean S1–S3 test; greedy vs sampled σ	amendment to family S, before seed-1/2 data
D64	10-17 21:57	The label "anti-propagation" for a CI below 0	amendment, before any A6 model result
D67	10-18 00:06	The bf16 merge was a bug: corrected (unmerged, 4-bit) evals become PRIMARY for families S, N, E; merged results kept, labelled superseded	amendment, before any corrected result
D69	10-18 09:48	Post-hoc baselines re-fitted on 4-bit SFT calibration completions	amendment, before any such completion
D71	10-18 09:50	Review round 2, R2-1–R2-7: eval hygiene (reload fingerprint), SFT revalidation, seed-level statistics (R2-4), uniform-credit seeds (R2-6), …	amendment, before any result it affects
D77	10-19 10:31	R2-1: below the log-prob floor, the σ-token criterion decides; the check is enforced	amendment (future fingerprints only)
D91	10-20 23:22	X4: a direct mode, a perturbed control, the cluster bootstrap, token cost	amendment, before any X4 data
D92	10-20 23:38	X2 robustness checks, defined before computing	analysis plan, **after** X2's result
D93	10-20 23:45	Review round 3, R3-1–R3-10: X4 additions; X2 robustness; optimization pilots; Phase 12 scope = aleatoric propagation (R3-4); A6c (R3-5); …	amendment, before any result it affects
D112	10-25 17:13	Review round 4, R4-1–R4-9: D-pos-v2 vs B-v2 on mixed-level data, MC-A, the frozen set, framing (R4-8)	amendment, before any result it affects
D114	10-25 17:31	Two deviations found while building: eval budget 1,280 tokens; 6 rows dropped	deviation, before any R4 output
D116	10-25 19:07	D-pos-v2's eval OOM'd in its backoff; fixed and rerun at batch 16	deviation, before any completion was scored
D122	10-26 07:52	MC-B: two deviations found while building	deviation, before any output
D125	10-26 08:34	D118 §6 amended: the primary P is D113's full criterion (K1 $\wedge$ C2 $\wedge$ C3)	amendment, before any confirmatory result
D131	10-28 02:52	D-pos-v2 seed 1's eval hung after a GPU driver reset; rerun on identical code	deviation (operational), before the analysis
D133	10-28 10:04	Review round 5, R5-1–R5-10: framing; tail autopsy (R5-2); hybrid oracle (R5-3); RSS fallback (R5-4); cross-model replication (R5-5); cuts (R5-7); …	amendment, **after** #32 / #33, before anything it asks for

entry	date	what	timing
D135	10-28 10:24	R5-4's RSS-fallback bounds, $\sigma/\mathrm{RSS} \in [0.333, 5]$, chosen on the development sets and frozen before one application to the confirmatory logs	analysis plan; grid designed **after** seeing #35
D138	10-28 18:46	R2-1's fingerprint for the Llama base records at sampling batch 4	deviation (implementation), before any pilot output
D139	10-29 00:13	D137's go rule runs as soon as the D-pos-v2 eval exists (same code, same logs); SFT seed 1 only on GO	amendment (timing), before the go rule runs

Every amendment and deviation in the decision timeline (dates IST, 2026), with when it was made relative to the data it affects. Source: docs/decision_timeline.md.

6 Optimization: σ rides on the answer's advantage

[To be written (docs/paper_outline.md, section 4; FINDINGS #6, #14, #16, #18, #25).]

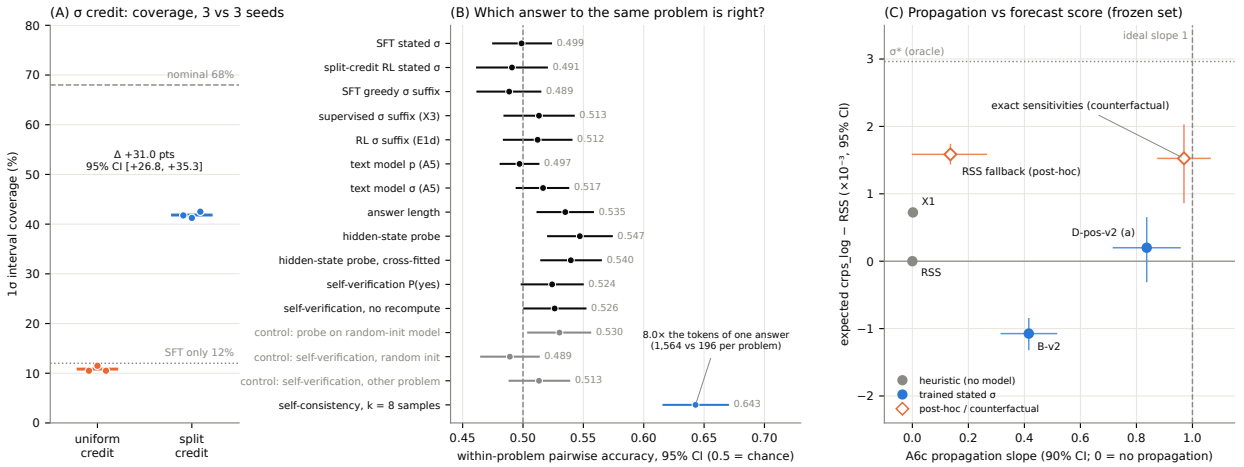


Figure 1: The three findings. (A) Uniform versus split credit: 1σ coverage per seed. (B) Within-problem pairwise accuracy of every correctness readout measured, against self-consistency. (C) On the frozen set, the noise-allocation slope against the expected-score difference from RSS. Every point is read from a committed result file; the sidecar `headline_data.json` traces each one.

7 Does the model know which of its answers is right?

[To be written (docs/paper_outline.md, section 5; FINDINGS #7, #19–#22, #26, #37).]

Table 3: Correctness signals against their inference cost.

readout	pairwise accuracy [95% CI]	tokens / problem
self-consistency (agreement of the other samples)	0.643 [0.616, 0.671]	1,564
hidden-state probe	0.547 [0.520, 0.574]	196
the same probe, random-init model	0.530 [0.503, 0.557]	196
completion length	0.535 [0.511, 0.559]	196
self-verification P(yes) (X4)	0.524 [0.498, 0.550]	196
stated σ (SFT)	0.499 [0.474, 0.524]	196

Within-problem pairwise accuracy (a right answer ranked above a wrong one to the same problem; 516 problems, 95% CIs clustered over problems) against generated tokens per problem (k = 8 samples for self-consistency). Source: results/r5_diag/cost.json (FINDINGS #22, #37).

Table 4: MC-A's spread against its cost, and MC-B's failure split.

signal	AUROC	extra tokens / problem
$\sigma^\star$	0.665	0
RSS	0.654	0
X1	0.666	0
B-v2 stated σ	0.655	0
MC-A raw spread, $k = 2$	0.638	496
MC-A raw spread, $k = 4$	0.710	992
MC-A raw spread, $k = 8$	0.745	1,984

A6c (development), AUROC of the signal for B-v2's run-0 answer being wrong, against the extra generated tokens per problem. MC-B's programs on the same set: 79.7 % fail at the interface, 5.1 % at run time, 14.7 % run but compute the wrong chain, 0.5 % are correct. Source: results/r5_diag/cost.json (FINDINGS #31, #33, #37).

8 Propagation of measurement noise: structure is not forecast quality

[To be written (docs/paper_outline.md, section 6; FINDINGS #17, #23, #27–#39).]

Table 5: The R4-3 confirmatory run: D-pos-v2 and B-v2 on the frozen set.

test	statistic	estimate [90% CI]	p	Holm
K1	D-pos-v2 (a): A6c slope > 0 (primary)	$+0.816$ [$+0.707$, $+0.923$]	$<1 \times 10^{-4}$	rejected
K2	D-pos-v2 (b) line's X: A6c slope > 0	$+0.884$ [$+0.745$, $+1.041$]	$<1 \times 10^{-4}$	rejected
K3	D-pos-v2 (c) exact from terms: A6c slope > 0	$+0.843$ [$+0.713$, $+0.987$]	$<1 \times 10^{-4}$	rejected
K4	B-v2: A6c slope > 0	$+0.416$ [$+0.315$, $+0.517$]	$<1 \times 10^{-4}$	rejected
K5	D-pos-v2 − B-v2 (expected crps_log)	$+0.0013$ [$+0.0008$, $+0.0017$]	$<1 \times 10^{-4}$	rejected
K6	D-pos-v2 − RSS	$+0.0002$ [-0.0002, $+0.0006$]	0.201	–
K7	D-pos-v2 − X1	-0.0005 [-0.0009, -0.0001]	0.991	–
K8	B-v2 − RSS	-0.0011 [-0.0013, -0.0009]	1.000	–
C1–C3	D-pos-v2 (a): A6b slope $+0.036$ [-0.057, $+0.212$]	C1 yes, C2 yes, C3 yes		
C1–C3	D-pos-v2 (b): A6b slope -0.026 [-0.049, -0.007]	C1 yes, C2 yes, C3 yes		
C1–C3	D-pos-v2 (c): A6b slope -0.023 [-0.047, -0.005]	C1 yes, C2 yes, C3 yes		
C1–C3	B-v2 (a): A6b slope $+0.003$ [-0.086, $+0.129$]	C1 yes, C2 no, C3 yes		

Seed means over 3 seeds per arm, frozen R4-5 set (3,060 problems); stage B's joint pairs bootstrap, $B = 10{,}000$; 90% CIs; one-sided p; Holm over K1–K8 at 0.05. Expected-score rows: oracle-centred expected crps_log, pooled over the three sets, clustered over pairs. P = K1 ∧ C2 ∧ C3 (D125): confirmed. Source: results/stage_r4/confirmatory/results.json (D118, D125; FINDINGS #32).

Table 6: Tail autopsy: what the worst problems get wrong.

primary cause	problem-seeds	share
sensitivity: value-dependent: formula	177	39 %
sensitivity: value-dependent: substitution (own wrong intermediates)	87	19 %
copy: decimal shift	71	16 %
chain selection: wrong number of terms	35	8 %
sensitivity: power-law exponent	28	6 %
format: no answer σ	26	6 %
sensitivity: value-dependent: formula (no expression)	18	4 %
format: no parsable σ line	7	2 %
aggregation	5	1 %
sensitivity: transcription (S line right, σ-line coefficient wrong)	1	0 %
sensitivity: value-dependent: substitution (values)	1	0 %

D-pos-v2 (a), frozen R4-5 set, 3 seeds; the worst 5% of problems by regret against $\sigma^\star$ per seed (456 problem-seeds), classified by primary cause. They hold 69 %–83 % of each seed's regret. Descriptive (R5-2, D133). Source: results/r5_diag/tail.json (FINDINGS #35).

Table 7: Component swaps: which part of the explicit σ costs the score.

σ with	− RSS [90% CI]	gap closed	A6c slope
(a) as stated	+0.0002 [−0.0002, +0.0006]	7 %	+0.837
exact sensitivities (model's levels, aggregation, copy)	+0.0015 [+0.0010, +0.0020]	51 %	+0.970
exact levels (model's sensitivities, aggregation, copy)	+0.0002 [−0.0002, +0.0006]	7 %	+0.838
exact aggregation (model's terms and copy)	−0.0008 [−0.0016, −0.0001]	−28 %	+0.806
exact copy, i.e. σ = the line's X (= (b))	+0.0002 [−0.0003, +0.0007]	8 %	+0.857
model's terms, exact aggregation and copy (= (c))	−0.0018 [−0.0040, −0.0002]	−60 %	+0.826
fully exact: the builder's line, unrounded (ceiling)	+0.0030 [+0.0028, +0.0031]	100 %	+0.998

One component of D-pos-v2's σ replaced by its exact value at a time (R5-3, D133): a diagnostic of where the loss comes from, never a method. Slopes on every A6c pair, an unanswered member at the floor. RSS-to-$\sigma^\star$ gap 0.00296 (expected crps_log). Source: results/r5_diag/tail.json (FINDINGS #35).

Table 8: The cross-model pilot on Llama-3.2-3B-Instruct.

item	statistic	value	met
K1, C2	D-pos-v2 (a): A6c slope	+0.785 [+0.595, +0.974] (505 pairs)	yes, yes
C3	D-pos-v2 (a): A6b slope, computation-correct pairs	4 pairs: not measurable	no
go	K1 ∧ C2 ∧ C3 (D137)		NO-GO
desc.	A6b slope, all answered pairs (Llama / Qwen s0)	−0.168 [−0.220, −0.119] / +0.010	
desc.	B-v2 (a): A6c slope	+0.396 [+0.246, +0.544]	
desc.	D-pos-v2 − B-v2 (expected crps_log)	−0.0089 [−0.0170, −0.0019]	
format	D-pos-v2: parsed / units / computation-correct	99.8 % / 99.7 % / 8.0 %	
format	B-v2: parsed / units / computation-correct	100.0 % / 100.0 % / 8.0 %	

Seed 0, frozen A6c + A6b (D137, D146). The go rule is R4-3's criterion on D-pos-v2's answer σ; C3 needs at least three computation-correct A6b pairs. Descriptive rows are D137's, outside the go rule. Sources: results/stage_xm/pilot/results.json, results/stage_xm/go_dpos2/go.json (FINDINGS #38).

Table 9: R5-6 part J: Jacobians outside the training range of trained chains.

test	statistic	estimate [90% CI]	p	Holm
J1	D-pos-v2 (a): J-sensitivity slope > 0 (primary)	+0.985 [+0.868, +1.103]	$<1 \times 10^{-4}$	rejected
J2	explicit − implicit slope	+0.530 [+0.377, +0.676]	$<1 \times 10^{-4}$	rejected
J3	D-pos-v2 − RSS (expected crps_log)	+0.0011 [−0.0002, +0.0020]	0.072	–
J4	D-pos-v2 − X1	+0.0000 [−0.0013, +0.0008]	0.442	–
J5	B-v2 − RSS	−0.0005 [−0.0008, −0.0002]	0.997	–
J6	B-v2 − X1	−0.0016 [−0.0019, −0.0013]	1.000	–
source	D-pos-v2 (a)	sens. +0.985 [+0.868, +1.103]; inv. +0.026 [−0.055, +0.119]	C1–C3: yes	
source	D-pos-v2 (b)	sens. +1.007 [+0.861, +1.172]; inv. +0.026 [−0.028, +0.078]	C1–C3: yes	
source	D-pos-v2 (c)	sens. +0.980 [+0.835, +1.143]; inv. +0.004 [−0.081, +0.082]	C1–C3: yes	
source	B-v2 (a)	sens. +0.455 [+0.348, +0.564]; inv. +0.184 [−0.032, +0.400]	C1–C3: no	
comparator	$\sigma^\star$	sens. +1.000 [+1.000, +1.000]		
comparator	RSS	sens. +0.000 [+0.000, +0.000]		
comparator	X1	sens. −0.001 [−0.003, +0.002]		
comparator	clamped to the training range	sens. +0.831 [+0.737, +0.919]		

Seed means over 3 seeds per arm on the frozen J set (1,632 problems); bootstrap as Table 5; Holm over J1–J6. C3 uses computation-correct invariance pairs. P_J = J1 ∧ C2 ∧ C3 (D147): confirmed. The clamped comparator propagates exactly but with every sensitivity clipped into its chain's training range. Source: results/r5_6/results.json (D147; FINDINGS #39).

Table 10: R5-6 part J: J-sensitivity slopes where clamping matters most.

subgroup	pairs	D-pos-v2 (a)	B-v2 (a)	clamped
exceedance below the median	408	+1.09 [+0.96, +1.22]	+0.53 [+0.37, +0.69]	+0.99 [+0.91, +1.06]
exceedance above the median	408	+0.83 [+0.63, +1.04]	+0.36 [+0.22, +0.49]	+0.62 [+0.45, +0.78]
strict	255	+0.89 [+0.65, +1.12]	+0.39 [+0.09, +0.72]	+0.71 [+0.57, +0.85]
rank flip	43	+1.12 [+0.88, +1.38]	+0.18 [−0.13, +0.41]	+0.36 [+0.00, +0.80]

Descriptive, outside the Holm family (D147 §10). Pairs: J-sensitivity and J-invariance pairs in the subgroup; slopes on the J-sensitivity pairs. Strict: also outside the range over every training chain to the same target. Rank flip: the dominant given was never dominant for that chain in training. Source: results/r5_6/results.json.

9 Discussion

[To be written (docs/paper_outline.md, section 7).]

10 Limitations

[To be written (docs/paper_outline.md, section 8): one model family; a synthetic domain; single-seed pilots; fresh Jacobians only (D148); the Llama pilot's descriptive invariance slope.]

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A Pre-registration ledger

[Every pre-registered hypothesis marked supported, not supported, or not run (`docs/decision_timeline.md`).]

B Reproducibility

[Reload and provenance checks, pinned bases, the bf16 merge pitfall, frozen-set checksums, compute (docs/paper_outline.md, appendix B). Every generated value, table and figure lists its committed sources in `paper/generated/SOURCES.md`.]